

VidEngram: Agentic Long-Term Video Understanding via Structured Hierarchical Memory

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Abstract

Long-video question answering is wasteful and lossy when a multimodal model must re-encode the same video for every query: context limits force sparse sampling or truncation, and no reusable representation remains for later questions. We present **VidEngram**, an agentic video QA system that builds a persistent hierarchical memory once at ingest and answers later questions from that memory. VidEngram captions video and audio jointly with Qwen2.5-Omni, consolidates segment memories into episode and entity layers, and uses a text-only ReAct planner with specialized retrieval tools to produce grounded answers. The web interface makes this process inspectable: users can watch memory formation, inspect retrieval traces, verify cited evidence in the video player, and explore entity relationships. On the full OmniVideoBench, VidEngram with a gpt-5-mini planner outperforms a strong single-pass Qwen2.5-Omni baseline (37.0% vs. 32.3%), with gains concentrated on longer videos where single-pass coverage becomes sparse. Because perception is paid once, subsequent queries use low-cost text-only planning. VidEngram is available as a hosted demo, walkthrough, and open-source code.

1 Introduction

Long-video question answering is a repeated-query problem: the same lecture, documentary, or meeting may be queried many times, but a multimodal model typically re-encodes the video for every question and compresses it into a bounded context window. This makes single-pass QA expensive and lossy: sparse frame sampling breaks temporal continuity (Zhang et al., 2023), flat RAG retrieves isolated caption snippets without narrative structure (Gia et al., 2025), and even long-context models (Gemini Team, Google, 2024) provide no reusable memory. The bottleneck is therefore also representational: users need an inspectable memory that can be reused across questions.

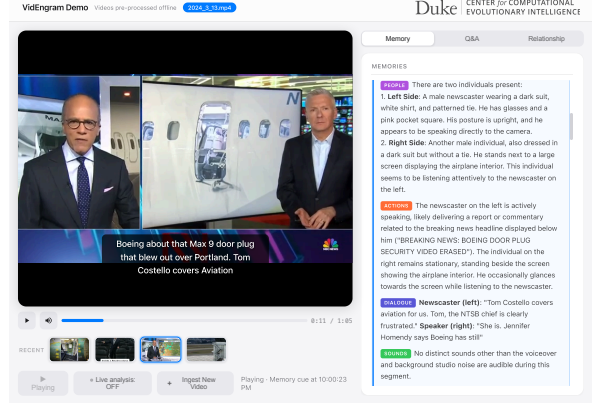


Figure 1: Memory formation view. The user watches the source video while the eleven-field structured-caption feed is populated in real time, making the stored memory inspectable before later questions are asked.

We present **VidEngram** (Figure 1), which treats video understanding as memory formation and retrieval. Rather than compressing a video into one context window per query, VidEngram encodes it once into a structured, indexed memory; at query time, a ReAct agent selects retrieval tools over that memory and grounds answers in retrieved evidence. In the web interface, users can watch memory formation, inspect retrieval traces, jump to cited video evidence, and explore entity relationships.

Contributions.

- Reusable video memory.** VidEngram converts video and audio into structured segment memories and a segment $\rightarrow$ episode $\rightarrow$ entity hierarchy.
- Inspectable agentic retrieval.** A ReAct loop (Yao et al., 2023) selects among memory and clip-verification tools, exposing retrieval traces and evidence for user inspection.
- Decoupling perception from reasoning.** The omni-model runs once at ingest; each query

uses a swappable text-only planner over memory, so planner quality can improve without re-encoding video.

2 Related Work

Video understanding and hierarchical memory.

General-purpose vision-language models (Radford et al., 2021; Alayrac et al., 2022; Li et al., 2023) and video extensions (Zhang et al., 2023; Maaz et al., 2024; Lin et al., 2024; Wang et al., 2024a,b) perform well on short clips, while long-context variants (Zhang et al., 2024b; Shen et al., 2024) extend the input window but still degrade with duration. Agentic and memory-based systems address this pressure differently: VideoAgent (Fan et al., 2024) adds tool use without a persistent backend, VideoARM (Yin et al., 2025) builds hierarchical memory at query time through coarse-to-fine backtracking, and HippoMM (Lin et al., 2025) introduces an episodic audiovisual hierarchy. VidEngram instead consolidates a persistent segment/episode/entity store once at ingest, serves it through keyword and vector indices with cross-video scope, and uses a single Qwen2.5-Omni pass rather than separate visual and speech encoders. Related systems build compressed or hierarchical memories (Zuo et al., 2025; Yuan et al., 2025; Wang et al., 2025b; Song et al., 2024; He et al., 2024) or retrieve over video (Ataallah et al., 2024; Zhang et al., 2024a), but do not combine a symbolic episode/entity store, persistent indexing, and agentic timestamped retrieval. Benchmarks for this setting include Video-MME, LVBench, EgoSchema, MLVU, NExT-QA, LongVideoBench, and OmniVideoBench (Fu et al., 2025; Wang et al., 2025a; Mangalam et al., 2023; Zhou et al., 2025; Xiao et al., 2021; Wu et al., 2024; Li et al., 2025).

Retrieval-augmented generation and agentic reasoning. Retrieval-augmented generation (Lewis et al., 2020) with dense retrievers (Karpukhin et al., 2020) has also been adapted to video QA. Video-RAG (Luo et al., 2024) injects visually aligned auxiliary text into an LVLM in a single pass; VRAG (Gia et al., 2025), RAVU (Malik et al., 2025), and VideoRAG (Ren et al., 2025) add retrieval or graph reasoning over long videos. These systems remain largely single-pass over flat or graph stores, without LLM-guided narrative consolidation, full-timeline entity resolution, or an inspectable agentic retrieval

loop. VidEngram instead applies ReAct (Yao et al., 2023) tool use (Schick et al., 2023) to persistent structured video memory.

3 Method

3.1 System Overview

VidEngram separates video QA into offline memory formation and online memory retrieval (Figure 2). During ingest, the system segments a video, captions each segment with Qwen2.5-Omni, consolidates captions into segment, episode, and entity layers, and writes these records to a persistent memory manager (Hu et al., 2026). At query time, a text-only ReAct planner searches this memory through specialized tools and returns an answer grounded in retrieved timestamped evidence.

We use the memory manager only as a persistent storage and hybrid-retrieval substrate; VidEngram constructs the video-specific memories, controls the retrieval tools, and handles timestamp-grounded answer presentation.

The web interface exposes the same pipeline through Memory, Q&A, and Relationship panels (§4), while a command-line interface supports batch ingest, scripted evaluation, and reproducible queries.

3.2 Temporal Segmentation

Temporal segmentation defines the unit of memory formation. When Whisper (Radford et al., 2023) is enabled (the preferred mode), the video is segmented at ASR sentence boundaries; otherwise the system falls back to ffmpeg-based silence and scene-change detection. Across our benchmark videos, this yields approximately 7.7 segments per minute, with an average segment length of 7–8 seconds. The short segment length is deliberate: it keeps captioning local enough for precise timestamp citations, while later episode consolidation restores broader narrative context for retrieval.

3.3 Unified Audiovisual Encoding

Each segment is encoded as a structured audiovisual memory using Qwen2.5-Omni-7B (Xu et al., 2025). Unlike systems such as HippoMM (Lin et al., 2025) that run separate visual and speech encoders followed by fusion, Qwen2.5-Omni processes frames and audio in a single forward pass. This lets the captioner describe visual events, speech, non-speech sounds, and their temporal

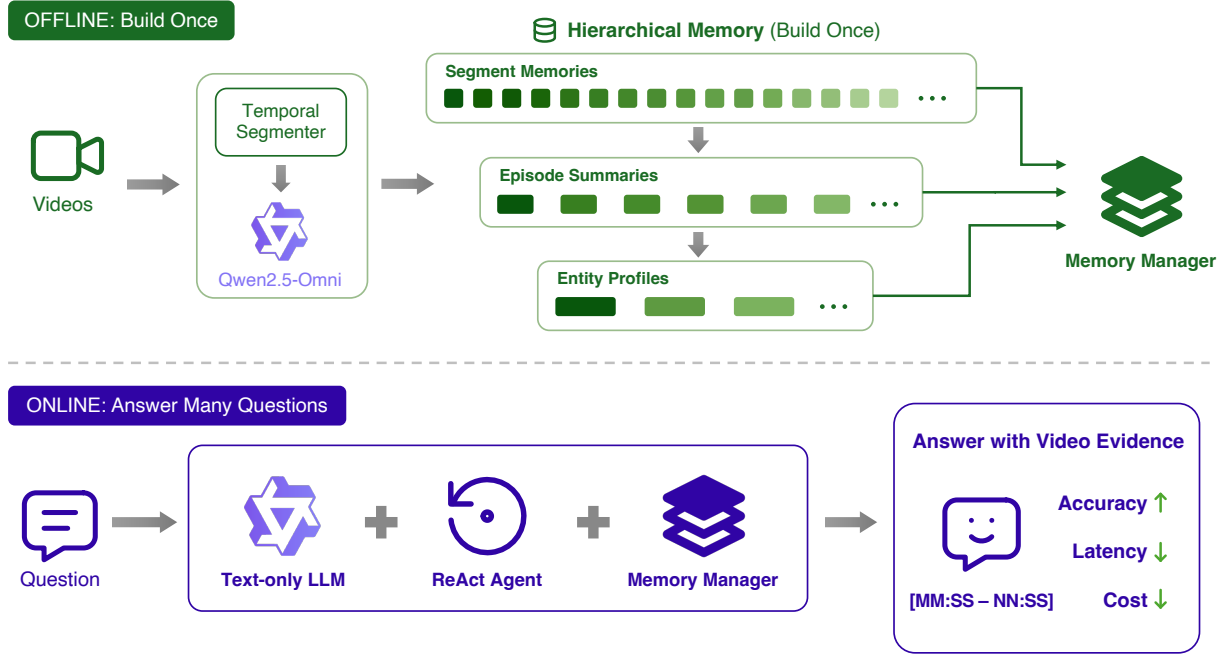


Figure 2: VidEngram architecture. **Offline (build once)**: each video is segmented, captioned by Qwen2.5-Omni into eleven structured fields, and consolidated into a three-layer memory (segment → episode → entity) in the memory manager. **Online (answer many)**: a ReAct agent answers each question from memory through specialized retrieval tools and timestamp citations, so users can inspect how an answer was produced without re-encoding the full video.

alignment without requiring a separate audiovisual fusion or diarization module.

Captions use an eleven-field template: SCENE, PEOPLE, ACTIONS, DIALOGUE, SOUNDS, AV-SYNC, TEXT, OBJECTS, COUNTS, EMOTION, and TEMPORAL. When Whisper ASR is available, its verbatim transcript is injected as a hint for DIALOGUE, letting Qwen2.5-Omni focus on attribution and audiovisual alignment.

These fields are stored as text rather than collapsed into a single free-form caption. This preserves different evidence types for later retrieval: exact dialogue can be found by keyword search, people and appearances can feed entity profiles, and visual events can be consolidated into episode-level memories.

3.4 Three-Layer Memory Consolidation

Caption consolidation turns local segment descriptions into a searchable video memory. VidEngram stores three memory layers independently: **Segment** memories preserve the eleven-field caption and ASR transcript for each short clip; **Episode** memories summarize contiguous segments into coherent narrative units; and **Entity profiles** merge repeated mentions into canonical people or objects with aliases, roles, and appearance timelines.

Stage 1: Segment deduplication. Adjacent captions above a Jaccard word-set threshold (default 0.7) are merged into one extended segment. This reduces redundant near-identical memories in static or slow-moving scenes while preserving timestamp coverage.

Stage 2: Episode grouping. Contiguous segments are grouped into episodes, each covering a coherent topic or scene with a start and end time. After each new segment, an LLM decides whether to extend the current episode or open a new one, subject to a maximum segment count, and summarizes each episode into a 3–5 sentence narrative for retrieval.

Stage 3: Entity resolution. Entity profiles are built in two steps: the LLM first extracts episode-level mentions with names, roles, and time ranges, then merges these mentions into canonical profiles with aliases, appearance timelines, and holistic descriptions. This provides an explicit entity layer for retrieval and relationship inspection, rather than relying only on a flat list of captions.

The resulting segment, episode, and entity records are written to the memory manager, tagged by layer, and indexed for hybrid retrieval (§3.1). A representative 60-minute video produces 483 segment memories, 79 episode summaries, and 14 en-

Tool	Retrieval	Best for
search_episodes	Hybrid (BM25+vec+RRF)	Factual / visual queries
search_speech	BM25 keyword	Exact word / quote lookup
search_profiles	Hybrid over entity	Who-is / appearance
search_deep	LLM-guided multi-hop	Complex reasoning
look_at_clip	Qwen2.5-Omni re-anal.	Audio-visual verification
get_timeline	Chronological listing	Time-range summaries

Table 1: The six tools available to the VidEngram ReAct agent. Hybrid tools use BM25/vector fusion; look_at_clip verifies a short clip when memory is insufficient.

tivity profiles. Segment memories preserve local evidence and timestamps, episode memories reduce retrieval fragmentation for scene-level questions, and entity profiles collect repeated appearances or mentions across the full timeline.

3.5 Streaming Ingest Architecture

The ingest pipeline is designed to make memory formation visible during the demo. VidEngram first runs ASR to obtain sentence boundaries and transcript hints, then submits the resulting segments to Qwen2.5-Omni for captioning in parallel. Completed captions are immediately written and indexed as segment memories, so users can watch the memory store grow rather than waiting for a batch output. Episode and entity consolidation run incrementally over this stream. The write rate is bounded mainly by Omni-model parallelism, such as served workers or API concurrency. On a 60-minute video, the first segment memory is ready in ~ 8 s, versus ~ 16 min if all captions must finish before memory is written.

3.6 ReAct Orchestrator

At query time, VidEngram answers from memory with a text-only planning LLM in a ReAct loop (Figure 3). Given the question, dialogue context, video metadata, and tool specification, the planner first issues a retrieval action, then answers only after observing sufficient retrieved evidence.

THINK AND ACTION. At each iteration, the planner identifies the missing evidence and selects an appropriate retrieval tool. Table 1 lists the available tools and their intended use cases. The tools retrieve from pre-built segment, episode, and entity memories. The only exception is look_at_clip, which calls Qwen2.5-Omni only on the final fallback iteration, after memory retrieval has failed to provide sufficient evidence. In that case, memory retrieval first narrows the target time span, so Qwen2.5-Omni re-examines only a short candidate

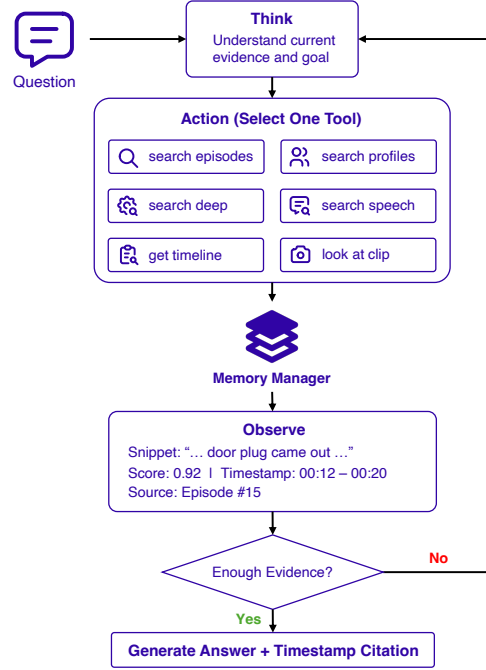


Figure 3: ReAct retrieval workflow. The planner selects among six tools; the memory manager returns evidence, and the agent iterates until it has enough support to answer with a timestamp citation.

clip rather than the full video.

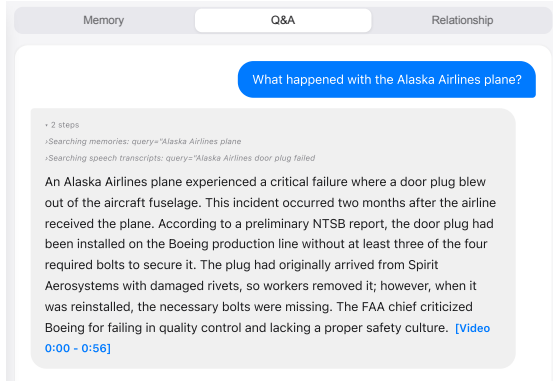
OBSERVE. Each tool returns ranked snippets with metadata and canonical timestamps. Observations are appended to the planner context, where scores rank evidence and trigger fallbacks but are not treated as calibrated confidence. The planner then either continues the loop or generates an answer grounded in the collected evidence. Final answers retain only citations traceable to segment metadata, so users can inspect the cited video span. The public setting allows up to five iterations before answering from the strongest observations.

3.7 Cross-Video Analysis

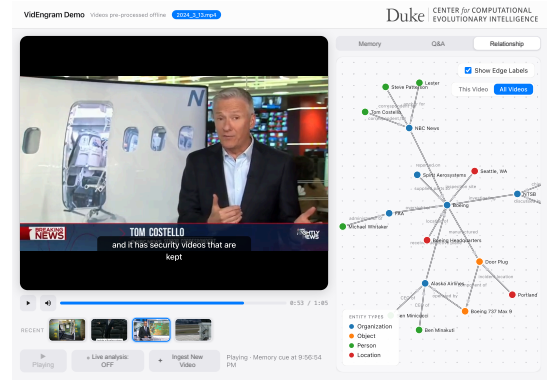
The memory manager’s group-id filtering lets the agent search all ingested videos simultaneously without additional infrastructure. In “all videos” mode, timestamp citations are suppressed for cross-video results, since memories from different videos do not share a common timeline. The same interface also supports collection-level relationship analysis by merging entity profiles across the ingested videos.

4 Demonstration

VidEngram is shown as a browser-based demo (Figures 1 and 4) organized into Memory, Q&A, and



(a) Q&A case study



(b) Relationship view

Figure 4: Demonstration views. **Left:** In the Q&A case study, a user asks what happened with the Alaska Airlines plane. VidEngram first searches video memories for the event, then searches speech transcripts for the more specific door-plug failure, and returns a timestamp-grounded answer whose citation can be clicked to verify the evidence in the video player. **Right:** In the Relationship view, the user inspects cross-video entity nodes, profiles, and source scenes.

Relationship panels. Attendees can inspect memory records, retrieval tool choices, returned evidence, and cited playback spans.

Memory. The Memory panel makes ingestion inspectable. It shows the eleven-field structured-caption feed beside time-synced playback (Figure 1). In the on-site demo and walkthrough, this feed fills as streaming ingest runs (§3.5); in the hosted Space, the same records are precomputed so visitors can inspect them without running the full captioning stack.

Q&A. The Q&A panel exposes the retrieval trace for each answer. In the case study in Figure 4, left, the question “What happened with the Alaska Airlines plane?” first triggers a broad episode-memory search and then a targeted speech search for the door-plug failure. The answer combines the two observations and cites [Video 0:00 - 0:56], allowing the user to verify the evidence in the player. This example illustrates why the agent uses multiple retrieval modes: the first search identifies the event context, while the speech search recovers the more specific phrase that supports the final answer.

Relationship. The Relationship panel exposes the entity layer. It renders extracted entities as an interactive graph (Figure 4, right). In the case study, the graph connects Boeing to the door plug, Alaska Airlines, Boeing Headquarters, FAA, Spirit AeroSystems, NBC News, Seattle, and NTSB, while labeling the relationships among these entities. Clicking a node opens its description and appearance timeline, linking the entity back to source

scenes for inspection in the demo. The same view can switch between a single video and merged cross-video memory, with cross-video timestamps suppressed when no common timeline exists.

Live demo.¹ The hosted Space uses preprocessed clips, so visitors can inspect real memory and retrieval traces without installing the system. The CPU-only version disables live ingestion and uses a lightweight BM25 reader, while the stored memory and agent remain unchanged; the on-site demonstration runs the full production pipeline. A walkthrough² and source code³ are available.

5 Evaluation

The evaluation tests two claims: reusable memory should remain robust as videos outgrow single-pass perception, and planner quality should affect accuracy without rebuilding memory.

5.1 Benchmark and Protocol

We evaluate on the full OmniVideoBench (Li et al., 2025): 628 videos and 1000 multiple-choice questions that couple audio and visual evidence. Scoring is exact letter match (A/B/C/D), with no LLM judge. Direct-Omni uses Qwen2.5-Omni-7B to answer each question in one pass over 256 uniformly sampled frames plus extracted audio, subject to the model processor’s audio limit. This is a strong single-pass recipe under our context and GPU budget. Both systems answer identical questions, so

¹<https://no00el-videngram-demo.hf.space/>

²<https://youtu.be/uhUpELRSkqo>

³<https://github.com/No00el/VidEngram>

Method	Planning LLM	Accuracy (%) by duration tier					Latency
		Short	Med.	Long	Ultra	Overall	med. (s)
Direct-Omni (Qwen2.5-Omni-7B)	—	48.5	29.9	26.0	30.7	32.3	7.0
VidEngram	Qwen2.5-7B	36.4	29.9	30.8	28.8	30.9	9.5
VidEngram	gpt-5-mini	45.5	37.5	33.5	34.1	37.0	6.4
VidEngram	gpt-5	45.5	35.8	33.5	36.7	37.1	16.5

Table 2: VidEngram on OmniVideoBench (1000 questions, 628 videos). Direct-Omni answers in one pass over 256 frames plus extracted audio; VidEngram captions once with the same Qwen2.5-Omni-7B model and varies only the text planner. gpt-5-mini reaches 37.0% (vs. 32.3%; McNemar $p = 0.012$), winning on medium, long, and ultra-long videos.

we use paired McNemar tests and report relative gains rather than leaderboard rank. For VidEngram, the Qwen2.5-Omni captioner is held fixed across all rows; only the text planner changes. This isolates the effect of planner choice at query time: the same stored memory can be reused with a local Qwen2.5 planner, gpt-5-mini, or gpt-5 without rebuilding the video index.

5.2 Results

VidEngram with a gpt-5-mini planner answers **37.0%** of the 1000 questions correctly, against **32.3%** for a strong single-pass Direct-Omni baseline (256 frames plus extracted audio), a statistically significant +4.7-point margin (paired McNemar $p = 0.012$). The advantage appears once a video exceeds what a single pass can cover (Table 2): VidEngram trails on short clips (−3.0), where one dense pass already sees and hears the whole clip, but leads on every longer tier (+7.6 medium, +7.5 long, +3.4 ultra-long). A fixed 256-frame budget becomes sparser as duration grows, while VidEngram answers from segment captions stored roughly every eight seconds plus the transcript. This pattern matches the intended use case: memory becomes useful as the same frame budget must cover longer timelines, where brief visual or spoken events are more likely to be missed by a capped single pass.

The planning LLM is the key lever. The planning LLM is a separate accuracy lever because it can be upgraded without rerunning the omni-model. Holding the stored memory and tool set fixed, VidEngram scores 30.9% with Qwen2.5-7B and 37.0% with gpt-5-mini. The gain saturates beyond gpt-5-mini: gpt-5 adds only +0.1 overall, so gpt-5-mini is the operating point.

Latency is insensitive to video duration. Each query retrieves from pre-built memory instead of re-encoding the video, so VidEngram’s median query latency stays near 6.4 s with the gpt-5-mini planner and changes little across duration tiers. The capped Direct-Omni baseline also keeps latency bounded, but only by fixing the input budget; as videos grow longer, the same cap makes visual and audio coverage increasingly sparse. Appendix C reports the latency breakdown by duration tier.

Cost is amortized across queries. VidEngram pays ingest once and reuses memory with low-cost text-only planning; re-encoding baselines pay video-processing cost per query. This favors VidEngram at high query volume. Appendix B gives the breakdown.

6 Conclusion

VidEngram demonstrates long-video QA as reusable, inspectable memory. A single omni-model ingest pass builds a persistent segment-episode-entity hierarchy, while later questions are answered by a text-only ReAct planner over specialized memory tools. The demo exposes both memory formation and retrieval traces, letting users inspect how answers are grounded in stored video evidence. On OmniVideoBench, this design helps most beyond short clips, where capped single-pass perception becomes sparse, and planner upgrades improve accuracy without rebuilding memory.

7 Limitations

We do not isolate individual components: accuracy is a system-level result, and the information-preservation claim is supported only indirectly (no same-system ablation). Accuracy is bounded by

the captioner: all memory derives from Qwen2.5-Omni-7B captions, and the system remains weak on counting and fine-grained perception. We use a single benchmark, and cross-video reasoning is not yet quantified.

Responsible use. VidEngram builds persistent, cross-video entity profiles with appearance timelines, which on identifiable people raise privacy, consent, and data-retention concerns and enable re-identification across unrelated footage. We intend it for consented, archival, or educational content, not surveillance.

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A Deployment Details

The hosted demo uses preprocessed clips so reviewers can inspect memory, Q&A, and relationship views without installing the full stack. The production system uses the same VidEngram pipeline with a persistent memory manager, a FastAPI backend, vLLM-served Qwen2.5-Omni-7B for captioning and clip grounding, and text-only models for planning, embedding, and reranking. The storage substrate provides document, keyword, vector, and cache services; VidEngram writes all video-derived memories and controls the retrieval tools described in §3.6.

B Cost Breakdown

Cost methodology. We instrument every model call (whether to a self-hosted Qwen model or a hosted GPT/Whisper API) to log its exact input/output token count and latency, so we report cost as measured token counts multiplied by each model’s published market price (Whisper billed per audio-minute). This token-count $\times$ unit-price accounting is hardware-independent and reproducible, unlike an amortized GPU cost, which for the self-hosted Qwen models is effectively zero. The per-query latencies in Table 2 use the same instrumentation.

VidEngram’s cost advantage appears along the query dimension. At ~ 500 – 1000 tokens per second of native 720p video, a one-hour video would produce ~ 3.6 M tokens ($113\times$ the 32K context limit), making full-video QA infeasible without severe downsampling. The capped Direct-Omni baseline therefore grows only until it reaches the 32K limit, at about $\approx \$0.009/\text{query}$; beyond that point it remains low-cost but loses coverage. Across seven clips (5 s–30 min), measured separately from §5, VidEngram’s ingest cost grows with duration, while per-query usage stays at ≈ 8 K text tokens. Thus $C_{\text{total}} = C_{\text{ingest}} + n \cdot c_q$, with $c_q \approx \$4 \times 10^{-4}/\text{query}$, is nearly flat in the number of questions once the memory is built (Table 3).

Table 3 uses self-hosted Qwen2.5-7B planning to show a deployable low-cost setting. The gpt-5-mini and gpt-5 planners in Table 2 are still text-only calls, so switching to them increases only the small c_q term rather than reintroducing multimodal video processing on every query.

Dur.	1 query		10 queries		100 queries	
	VG	BL	VG	BL	VG	BL
5 s	\$0.002	\$0.001	\$0.006	\$0.010	\$0.042	\$0.098
10 s	\$0.002	\$0.002	\$0.006	\$0.020	\$0.042	\$0.196
30 s	\$0.007	\$0.006	\$0.010	\$0.059	\$0.046	\$0.588
1 min	\$0.016	\$0.009	\$0.019	\$0.092	\$0.055	\$0.917
5 min	\$0.078	\$0.009	\$0.081	\$0.092	\$0.117	\$0.917
10 min	\$0.172	\$0.009	\$0.176	\$0.092	\$0.212	\$0.917
30 min	\$0.574	\$0.009	\$0.577	\$0.092	\$0.613	\$0.917

Table 3: Measured cost (USD) over n queries: VidEngram (VG, self-hosted Qwen2.5-7B planning) vs. a Direct-Omni baseline (BL). Below ~ 47 s, BL’s per-query cost grows with duration (5 s: \$0.001, 30 s: \$0.006); beyond that it saturates at Qwen2.5-Omni’s 32K context ($\approx \$0.009/\text{query}$). VG’s per-query agent cost is negligible ($\approx \$4 \times 10^{-4}$), so VG is flat in n ; its measured per-clip ingest grows with duration. At one query the one-time ingest dominates (longer clips favor BL), but VG’s flat amortization undercuts the baseline across all durations by ≈ 100 queries.

Method	Short	Med.	Long	Ultra	Overall
Direct-Omni	6.34	6.56	7.06	7.16	6.99
VidEngram + Qwen2.5-7B	8.65	9.67	9.73	9.54	9.49
VidEngram + gpt-5-mini	5.81	6.69	6.55	6.21	6.43
VidEngram + gpt-5	11.98	15.91	18.24	19.00	16.51

Table 4: Median query latency (seconds) by OmniVideoBench duration tier (Short: 165 questions; Medium: 344; Long: 227; Ultra: 264).

C Query Latency by Duration

Table 4 breaks down median query latency by OmniVideoBench duration tier. The same 1000 questions and official duration tiers used in Table 2 are used here. Direct-Omni remains bounded because it uses a capped 256-frame input, while VidEngram answers from pre-built memory; planner choice therefore dominates the remaining query-time variation.

D Captioning Prompt

The eleven-field structured caption (§3.3) is produced by the system prompt in Figure 5, sent to Qwen2.5-Omni-7B, followed in deployment by a worked example segment (omitted for space) and a user message instructing the model to emit all eleven fields at length.

You are a precise, EXHAUSTIVE video analyst. For each video segment output EXACTLY these labeled fields, each on its own line. Record EVERY detail -- exact counts and quantities, colors, on-screen text, spatial relationships, object attributes, and fine-grained actions. Downstream questions are detailed and counting-heavy and will depend on the specifics you capture here, so err strongly on the side of MORE detail. NEVER summarize -- ENUMERATE the specifics instead (e.g. "red cape, blue mask, yellow emblem"; "3 books and 2 red cups"; "upbeat piano with drums").

SCENE: [Physical setting, location, lighting, camera angle and movement, background elements. Be exhaustive.]

PEOPLE: [Every visible person: identity/name if known, clothing and colors, position, expression, posture. State how many there are, and note relative attributes (tallest/shortest, leftmost/rightmost, oldest/youngest, nearest/farthest) so later questions can disambiguate which person .]

ACTIONS: [What each person does, all interactions and movements, described step by step in order.]

DIALOGUE: [When an ASR transcript is provided, use it verbatim ; attribute each utterance to the visually identifiable speaker (e.g. "John:", "Woman in red:"), else "Speaker A:", "Speaker B:". Transcribe yourself if no ASR is provided. "None" if the segment is silent.]

SOUNDS: [Every non-speech sound in detail: animal/object sounds, music (genre, instruments, tempo, mood), sound effects, ambient noise. Count occurrences.]

AV_SYNC: [Audio-visual co-occurrence: for every notable sound, speech line, or music cue, state what is visually happening at that exact moment and who is involved (identified by distinguishing attributes such as tallest or red shirt), linking each audio event to the simultaneous visual with "->". "None" only if the segment is fully silent.]

TEXT: [ALL visible on-screen text, signs, titles, subtitles -- transcribe verbatim. "None" if absent.]

OBJECTS: [ALL notable objects, tools, props: their colors, quantities, positions, and any text on them.]

COUNTS: [Explicit counts of anything countable: number of people, repeated objects, occurrences of an event. Always give numbers when possible.]

EMOTION: [Emotional tone with specific descriptors and nuance .]

TEMPORAL: [Sequence of events within this segment in order: what happens first, what changes, how it ends.]

Figure 5: The captioning system prompt sent to Qwen2.5-Omni-7B (§3.3). Each field is one line; long descriptions wrap.